

STATSIG lecture 4

Optimal filtering & signal estimation

Discrete-time Wiener filter

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Model

Desired signal	$Y(n)$
Blurring filter	$G(f)$
Noise	$W(n)$
Observed noisy and blurred signal	$X(n)$
Optimal (Wiener) filter to be designed	$H(f)$
Estimated signal	$\hat{Y}(n)$

Task

Find optimal (Wiener) filter	$\{h(n), H(f)\}$
that minimizes the cost function	$MSE \stackrel{\text{def}}{=} E[(Y(n) - \hat{Y}(n))^2]$

FIR optimal Wiener filter

FIR filtering of observed signal $\hat{Y}(n) = \sum_{m=-M_1}^{M_2} h(m) X(n-m)$

Minimizing $MSE \stackrel{\text{def}}{=} E \left[\left(Y(n) - \hat{Y}(n) \right)^2 \right] = E \left[\left(Y(n) - \sum_{m=-M_1}^{M_2} h(m) X(n-m) \right)^2 \right]$
gives optimal filter coefficients $\mathbf{h}(n) = \mathbf{R}_{xx}^{-1} \mathbf{r}_{xy}$ (Wiener-Hopf equation) where

filter coefficients $\mathbf{h}(n) = [h(-M_1), \dots, h(M_2)]^T$

autocorrelation matrix $\mathbf{R}_{xx} = \begin{bmatrix} R_{xx}(0) & R_{xx}(1) & \cdots & R_{xx}(M_1 + M_2) \\ R_{xx}(1) & R_{xx}(0) & \cdots & \\ \vdots & & \ddots & R_{xx}(1) \\ R_{xx}(M_1 + M_2) & R_{xx}(1) & R_{xx}(0) & \end{bmatrix} \mathbf{y}$

crosscorrelation vector $\mathbf{r}_{xy} = [r_{xy}(-M_1), \dots, r_{xy}(M_2)]^T$

Performance $MSE_{min} = E \left[\left(Y(n) - \hat{Y}(n) \right)^2 \right]_{min} = R_{yy}(0) - \sum_{m=-M_1}^{M_2} h(m) R_{xy}(m)$
 $\leq E \left[\left(Y(n) - X(n) \right)^2 \right] = E[W(n)^2] = \sigma_w^2$

IIR optimal Wiener filter

IIR filtering of observed signal $\hat{Y}(n) = \sum_{-\infty}^{\infty} h(m) X(n - m)$

Minimizing $MSE \stackrel{\text{def}}{=} E \left[\left(Y(n) - \hat{Y}(n) \right)^2 \right] = E \left[\left(Y(n) - \sum_{-\infty}^{\infty} h(m) X(n - m) \right)^2 \right]$

gives

$$R_{xy}(j) = \sum_{-\infty}^{\infty} h(m) R_{xx}(j - m), \quad j \in \mathbb{Z} \quad \text{or}$$

$$R_{xy}(\tau) = h(\tau) * R_{xx}(\tau)$$

$$H(f) = \frac{S_{xy}(f)}{S_{xx}(f)}$$

filter coefficients

$$h(n) = IDTFT\{H(f)\}$$

If signal $Y(n)$ and noise $W(n)$ are uncorrelated (often assumed)

$$R_{xx}(\tau) = R_{yy}(\tau) + R_{ww}(\tau)$$

$$R_{xy}(\tau) = R_{yy}(\tau)$$

and

$$H(f) = \frac{S_{yy}(f)}{S_{yy}(f) + S_{ww}(f)} = \frac{1}{1 + \frac{1}{SNR(f)}}, \quad SNR(f) = \frac{S_{yy}(f)}{S_{ww}(f)}$$

Performance

$$MSE_{min} = R_{yy}(0) - \sum_{-\infty}^{\infty} h(m) R_{xy}(m) = \frac{1}{f_s} \int_{-f_s/2}^{f_s/2} \left(S_{yy}(f) - |H(f)|^2 S_{xx}(f) \right) df$$

IIR filter deconvolution

IIR filtering of observed signal $\hat{Y}(n) = \sum_{-\infty}^{\infty} h(m) X(n - m)$

Minimizing $MSE \stackrel{\text{def}}{=} E \left[\left(Y(n) - \hat{Y}(n) \right)^2 \right] = E \left[\left(Y(n) - \sum_{-\infty}^{\infty} h(m) X(n - m) \right)^2 \right]$
gives again $H(f) = \frac{S_{xy}(f)}{S_{xx}(f)}$

For complete model including blurring $G(f)$ is here found

$$\begin{aligned} S_{xx}(f) &= |G(f)|^2 S_{yy}(f) + S_{ww}(f) \\ R_{xy}(\tau) &= g(-\tau) * R_{yy}(\tau) \Rightarrow \\ S_{xy}(f) &= G^*(f) S_{yy}(f) \end{aligned}$$

Therefore

$$H(f) = \frac{G^*(f) S_{yy}(f)}{|G(f)|^2 S_{yy}(f) + S_{ww}(f)} = \frac{1}{G(f)} \frac{S_{yy}(f)}{S_{yy}(f) + \frac{S_{ww}(f)}{|G(f)|^2}} = \frac{1}{G(f)} \frac{1}{1 + \frac{1}{|G(f)|^2 SNR(f)}}$$

where again $SNR(f) = \frac{S_{yy}(f)}{S_{ww}(f)}$